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CA-SOFTWARE (CA-S)

ASSIGNMENT – 3

Trends and Cloud Services Overview - Write a three-paragraph report covering: 1) the benefits of serverless architecture, 2) the concept of Progressive Web Apps (PWAs), and 3) the role of AI and Machine Learning in software architecture. Then, in one paragraph, describe the cloud computing service models (SaaS, PaaS, IaaS) and their use cases

1 Serverless Architecture Benefits:

- Developers focus solely on code; infrastructure is managed by the cloud provider.
- Autoscaling, high availability, and reduced operational cost.
- Suitable for event-driven applications and irregular workloads (eg, AWS Lambda, Azure Functions)

2 Progressive Web Apps (PWA):

- Web applications with enhanced capabilities: offline access, push notifications and installability.
- Combine the reach of the web with the experience of native apps.
- Built using HTML, CSS, JS with service workers and manifest files.

3 AI/ML in software Architecture and Cloud Services:

- AI/ML models integrated into services (eg, ~~recommend~~ ^{recommendations} engines, chatbots).
- Use of SaaS (eg, Google AI Tools), PaaS (eg, Azure ML Studio), and IaaS (for model training and GPU workloads).
- Enhances automation, user experience, and decision-making capabilities in apps.